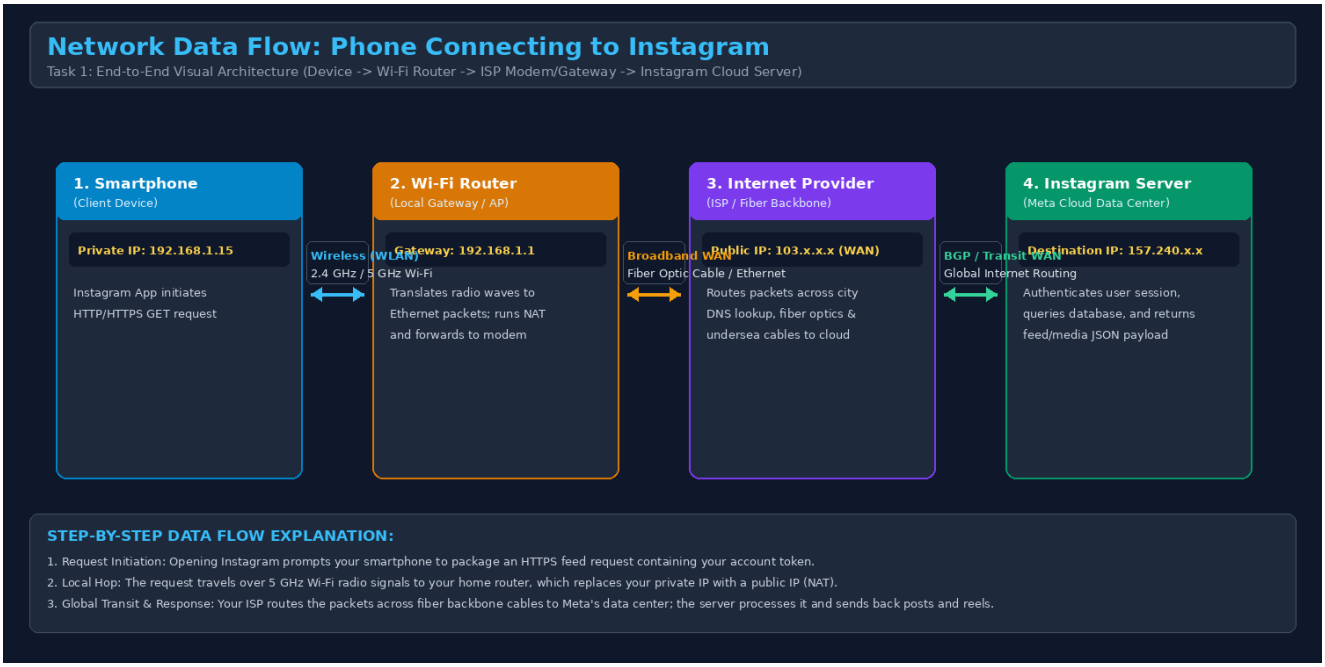
Introduction to Networks

Task 1: Diagram — Phone Connecting to Instagram

The diagram illustrates the path your request takes from your phone to Meta's data center:

- **Key Parts & Connections:**
    1. **Smartphone (Client):** Uses a private IP (e.g., 192.168.1.15) to initiate an encrypted HTTPS feed request.
    2. **Wireless Connection (WLAN):** 2.4 GHz or 5 GHz Wi-Fi radio frequencies link the phone to your local access point.
    3. **Wi-Fi Router (Gateway):** Translates radio waves to digital frames, assigns local IPs via DHCP, and performs Network Address Translation (NAT) to route traffic to the outside world.
    4. **ISP Broadband Network (WAN):** Fiber optic cables carry data packets across city nodes and regional Internet Service Provider routing stations.
    5. **Instagram / Meta Cloud Server:** Authenticates the session, retrieves updated posts and reels from its distributed database, and sends data back along the reverse path.



Task 2: 5 Everyday Networked Devices & Their Network Types

Device	Primary Network Type	Description of Connection
1. Smartwatch	**PAN (Personal Area Network)**	Connects directly to your nearby phone using short-range **Bluetooth** (≈ 10 meters) to sync steps, calls, and notifications.
2. Laptop / Desktop	**LAN (Local Area Network)**	Connects via Cat6 Ethernet cable or Wi-Fi to a router or office switch, sharing local printers, network shares, and internet access within a building.
3. Smart TV	**WLAN (Wireless Local Area Network)**	Connects to your home Wi-Fi router to stream high-definition video directly from local media servers or web streaming apps.
4. Wi-Fi Router / Gateway	**WAN (Wide Area Network)**	Its external (WAN) port interfaces directly with your ISP's regional network, bridging your private home network to the global internet.
5. Smartphone (on 4G/5G Cellular)	**WWAN / Cellular WAN**	Connects to cellular base tower antennas spanning miles across the city when disconnected from local home Wi-Fi.

Task 3: Difference Between the Internet and an Intranet (Real-World Example)

- **The Internet:**
    - **What it is:** The global, publicly accessible network of millions of interconnected servers, computers, and devices open to anyone worldwide.
    - **Example:** Using your college's public Wi-Fi to browse Google, stream YouTube, or access public websites. Anyone with an internet connection can view these services from anywhere on Earth.
- **An Intranet:**
    - **What it is:** A strictly private, secure internal network restricted only to authenticated members of a specific organization or institution. It is invisible and inaccessible to outsiders on the public web.
    - **Example:** Your college’s internal student portal, grade submission system, or offline campus library catalog. When connected to the college campus Wi-Fi, you can access internal IP resources (e.g., 10.0.0.50/portal), but if an outsider tries to open that address from their home internet connection, the page will fail to load.

Task 4: How Data Travels in Zomato When Searching for Food (4–5 Lines)

1. When you type *"Pizza"* and tap **Search** in Zomato, your phone packages this query into an encrypted HTTPS request containing your GPS location and account token.
2. The smartphone sends this data as 5 GHzWi-Fi radio waves to your home router, which uses NAT to attach your public IP address and forwards the packet through your ISP's fiber optic cables.
3. The ISP routes the packets across the Internet backbone directly to Zomato's cloud backend servers (hosted on AWS).
4. Zomato's database queries nearby restaurants in your area with pizza on their active menu, packages the results into a lightweight JSON payload, and routes it back through your ISP to your router.
5. Your router transmits the returning packets over Wi-Fi back to your phone, where the Zomato app immediately decodes the data and renders the list of nearby pizza places on your screen.